DO NOW — RECALL

Tree traversal: post-order and breadth-first

Name: _____ Class: _____ Date: _____

1. What does a constructor do?

2. What is in-order traversal order?

3. What does FIFO stand for?

4. A root R has only left child L. What is post-order?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Initialises an object when it is created.

2. Model answer

Left, root, right.

3. Model answer

First In, First Out.

4. Model answer

L then R.

Vocabulary — know these terms

post-order — Traverse left, right, then root.

breadth-first — Visit vertices level by level.

FIFO queue — First-in-first-out structure used by BFS.

leaf — Node without children.

traversal — Systematically visit each node in a structure.

NEW LEARNING

Read, annotate and explain

Depth-first post-order

Visit an entire left subtree, then right subtree, and finally the current node.

Breadth-first level order

Start with the root in a queue. Dequeue one node, visit it and enqueue each of its children.

Compare output

Same structure yields different orders: 2,6,4,12,8 versus 8,4,12,2,6.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Find post-order traversal of root 10, children 5 and 15, with 5 having left child 2.

Predict first

PostOrder(10) calls left(5), then left(2). At node2 both null-child calls return, so print2; return to node5 and print5.

Model the reasoning

Return to10, traverse right child15 and print15, then print10. Output 2,5,15,10: the print follows both child calls. Node has Key, Left and Right fields.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A preorder routine prints node.Key, then traverses Left and Right (with a null guard). Move one statement to make it post-order. Give the result for root6 with children3 and9.

Question 2

Tree root 12, left 5 with children 2 and 8, right 17. Give breadth-first and post-order traversals.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Write pseudocode to traverse a binary tree breadth-first. Explain why a queue rather than a stack is used.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Breadth-first traversal normally uses a stack to ensure earliest-discovered nodes are visited first.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.